



Dr B R Ambedkar National Institute of Technology Jalandhar Punjab
Department of Computer Science and Engineering
(Scheme of M. Tech. Computer Science and Engineering)

DETAILED CURRICULUM FOR M. TECH PROGRAMME
COMPUTER SCIENCE & ENGINEERING

CORE COURSES SEMESTER I

Department: Computer Science and Engineering					
Course Title: Algorithm Design and Analysis					Course Code: CS0501
Course Designation: Core					
Credit Scheme	L	T	P	C	
	3	0	0	3	
Course Assessment Methods: One sessional exam and one end-semester exam, along with assignments, presentations and class tests, which may be conducted by the course coordinator in lieu of internal assessment.					
Course Outcomes	The course will enable the student to:				
	CO1	Recall and interpret fundamental data structures and complexity notations; analyze their performance in the context of algorithm design paradigms.			
	CO2	Construct and evaluate hierarchical and multimedia data structures to solve domain-specific problems efficiently.			
	CO3	Analyze and compare advanced data structures like heaps, disjoint sets, and graph-based algorithms for real-world applications.			
	CO4	Design and analyze advanced algorithms and classify problems into complexity classes using polynomial-time reductions to establish NP-completeness.			
Topics to be covered:					
Fundamentals of Algorithm Design: Brushing up of Basic Data Structures, Order notations, Models of Computation: Space and Time complexity, Lower and upper bounds, Design paradigms: greedy, Divide-and-conquer, Dynamic programming, Branch and bound, and Backtracking techniques.					
Search and Multimedia Structures: Binary search trees, AVL trees, 2-3 trees, 2-3-4 trees, Red-black trees, B trees. Multimedia Structures like Segment trees, k-d trees, Point Quad trees, MX-Quad trees, R-trees, Splay Tree; Bags and Sets.					
Heaps and Graphs: Basic techniques, Min-max heaps, Deaps, Leftist heaps, Binomial heaps, Fibonacci heaps, Skew heaps analysis of Fibonacci heap and disjoint-set forest, Graph algorithms: Connectivity, Articulation Points, Topological sort, Minimum spanning trees, shortest paths, Network flow.					
Advanced Algorithms: String Matching Algorithms, Parallel Algorithms (The basics of fork-join parallelism, Parallel Matrix Multiplication, Parallel Merge Sort), Online Algorithms (Waiting for an elevator, Maintaining Search List, Online Caching), Linear Programming.					
Complexity Classes: Classes P and NP and Co-NP, reduction, NP-completeness, satisfiability problem (SAT), Subset sum problem, Vertex cover, Independent Set, Graph Coloring problem.					



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Text / Reference Books	1. Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, Fourth Edition, Pearson, 2013. 2. Adam Drozdek, Data Structures and Algorithms in C++, Fourth Edition, Cengage Learning, 2012. 3. Thomas H Cormen, Charles E Lieserson, Ronald L Rivest and Clifford Stein, Introduction to Algorithms, Fourth Edition, MIT Press/McGraw-Hill, 2022. 4. Sridhar S., Design and Analysis of Algorithms, Oxford university press, 2022. 5. Aditya Y. Bhargava, Grokking algorithms, Simon and Schuster, 2024.									
	CO-PO Mapping:									
	Course Outcomes	Program Outcomes								
					CSE		DSE		CSE (IS)	
		PO1	PO2	PO3	PSO1	PSO2	PSO1	PSO2	PSO1	PSO2
	CO1	1					1		1	2
CO2	2		2	2	2		1	3		
CO3	3	2	3		3	3			1	
CO4	3		3		3	3		2		
1 – Low, 2 – Medium, 3 - High										